

NAMAN YESHWANTH KUMAR

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SUMMARY

MS Computer Engineering (ASU, GPA 3.81, May 2026) working at the kernel/silicon boundary for AI inference. Wrote 6 CUDA kernels in C++ by hand for MLP training on MNIST — manual `cudaMalloc/cudaMemcpy`, `nvcc`, `ctypes` FFI — and modified `gem5`'s C++ source to add a first-class LRU-IPV cache replacement `SimObject`. Designed a Graph Neural Network ASIC accelerator end-to-end on ASAP7 7nm through Synopsys DC + Cadence Innovus + Voltus, so I understand both what the kernel runs on and what it's asked to compute.

TECHNICAL SKILLS

AI Kernels & CUDA: CUDA (hand-written kernels) | C++ | `nvcc` | `cudaMalloc/cudaMemcpy` | Python ↔ C++ FFI (`ctypes`) | Block/Thread Tuning | PyTorch | TensorFlow | NumPy

Computer Architecture & RTL: `gem5` (C++ `SimObject` internals) | 5-Stage Pipeline | Cache Hierarchy | SystemVerilog | Verilog | Cocotb | Synopsys Design Compiler | Cadence Innovus | Voltus | OpenROAD | Yosys | ASAP7 7nm | SkyWater 130nm

Languages & Tools: C/C++ | Python | Linux | Docker | Git | FastAPI | LangGraph | MCP Protocol | Multi-provider LLM Routing

EDUCATION

Master of Science — Computer Engineering (GPA 3.81/4.0)

Arizona State University

May 2026

Tempe, AZ

Coursework: Advanced Computer Architecture · GPU Architecture (CUDA) · Algorithm/Hardware Co-Design · SoC Design · VLSI Design

Bachelor of Engineering — Electrical and Electronics

BMS College of Engineering

August 2022

Bengaluru, India

PROJECT EXPERIENCE

CUDA MLP — Hand-Written Kernels for Neural Network Training

Fall 2024

- Implemented 6 CUDA kernels in C++ by hand for end-to-end MLP training on MNIST — **linear forward** (matmul + bias), **linear backward** (gradients w.r.t. inputs/weights/biases), and **ReLU forward/backward**; configured block/thread launch dimensions per kernel and managed host↔device transfers explicitly with manual `cudaMalloc/cudaMemcpy`.
- Compiled kernels via `nvcc` into shared libraries and bound them to Python through `ctypes` FFI for an end-to-end training pipeline that ran the full forward/backward/update loop from CUDA, not a framework — closing the loop from raw kernel code to a converged model on MNIST.

Graph Neural Network ASIC Accelerator — Full RTL-to-GDSII on ASAP7 7nm

Spring 2025

- Designed an 11-module GNN ASIC accelerator in SystemVerilog — 8 parallel MACs for 96-element dot products and a 12-state FSM for bidirectional COO edge aggregation — and completed the full RTL-to-GDSII flow: Synopsys Design Compiler synthesis on ASAP7 7nm RVT (1 ns clock, 10,878 cells), Cadence Innovus APR with CTS, and Voltus post-APR power analysis driven by VCD activity files.
- Achieved **50,300 μm^2** chip area at 50.6% core density, **52.29 mW** total power (64.8% switching), and **99.2 ns** end-to-end latency — full understanding of what an inference accelerator looks like at the silicon level.

LRU-IPV Cache Replacement Policy — `gem5` C++ Internals

Fall 2024

- Implemented LRU with Insertion Position Vector (IPV) replacement policy from scratch inside `gem5`'s C++ source as a first-class `SimObject`, integrating all 4 cache interface methods (`reset`, `touch`, `getVictim`, `invalidate`); modified 5 `gem5` source files and validated across 3 L2 cache configurations (128/256/512 kB), observing L2 replacements collapse from **637** → **112** → **2** as capacity grew — isolating clean capacity-vs-policy effects at 41.8M instructions, 2.04 IPC.

SiliconCrew — Autonomous LLM Agent for RTL-to-GDSII Design

Nov 2025 – Present

- Built an autonomous LangGraph agent (21 tools) that generates synthesizable Verilog from natural language, self-corrects via VCD waveform analysis on simulation failure, and synthesizes to GDSII through Yosys + OpenROAD; achieved **37/92 first-pass on CVDP** and **5/5 on ASU Spec2Tapeout**; multi-provider LLM routing (Gemini, OpenAI, Anthropic), full MCP server (`stdio`/SSE/HTTP) compatible with Claude Desktop.

5-Stage Pipelined MIPS Processor — SystemVerilog

Fall 2025 – Spring 2026

- Implemented a 5-stage pipelined MIPS processor in SystemVerilog (33 instructions) with a 12-path forwarding network (EX-to-EX, MEM-to-EX, WB-to-ID, branch, JR) and 6 hazard stall conditions; extended with an 8-entry fully-associative CAM instruction cache (3-state FIFO FSM) and LL/SC atomic operations with link-address tracking.

WORK EXPERIENCE

Systems Engineer

NCR GOLD

Oct 2022 – Apr 2024

Bengaluru, India

- Automated the entire order lifecycle for a wholesale business — production Django platform (PostgreSQL + Celery + Nginx) processing **18,000+ orders over 3 years**; reverse-engineered third-party ERP (zero public API) via SQL schema analysis for bidirectional sync, eliminating ~20 min of manual relay per order.

AWARDS & HONORS

- Won '**Best Use of AI**' at Strategy X DevHacks'25 (ASU) — built AdaptED AI, a Gemini-powered platform that generates personalized flashcards, quizzes, and learning paths from uploaded documents in a 24-hour hackathon.
- Ranked **Top 10 nationally** in e-Yantra Robotics Competition (IIT Bombay) — designed a self-balancing vehicle in CoppeliaSim, February 2022.
- Secured **2nd place at Hack SoDA 2024** (Amazon-sponsored, 24-hour) — built PassGen, a secure offline Chrome Extension for unique password generation.